CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

INTERIM PROJECT REPORT

Project Status: Ongoing

Submitted by: SWARNA CHOUDHARY

ABSTRACT

Credit card fraud is a significant challenge in the financial industry, resulting in financial losses for both customers and financial institutions. With the increasing use of digital payment systems, identifying fraudulent transactions has become increasingly important.

This project aims to develop a machine learning-based credit card fraud detection system capable of distinguishing between genuine and fraudulent transactions. The project uses a publicly available dataset from Kaggle containing 284,807 transactions.

The initial stages of the project involve dataset inspection, data cleaning, preprocessing, and exploratory data analysis (EDA). Following these stages, machine learning classification algorithms are being implemented to identify fraudulent transactions.

Currently, data cleaning, preliminary exploratory analysis, and feature scaling have been completed. Logistic Regression is being implemented as the initial classification model. Further model evaluation and comparison will be conducted in the upcoming stages.

1. INTRODUCTION

Credit card fraud refers to unauthorized transactions performed using another individual's credit card information. The growth of online banking, e-commerce, and digital payment platforms has increased the importance of reliable fraud detection mechanisms.

Traditional fraud detection techniques often rely on predefined rules. However, machine learning algorithms can analyze historical transaction patterns and identify potentially

fraudulent activities. This project focuses on analyzing credit card transaction data and developing a classification model using machine learning techniques.

1.1 Problem Statement

The major challenge in credit card fraud detection is identifying fraudulent transactions among a significantly larger number of genuine transactions.

Since fraudulent transactions represent a very small percentage of the overall dataset, developing an accurate classification model requires appropriate data preprocessing and evaluation techniques.

1.2 Objectives

- To analyze credit card transaction data.
- To identify patterns in genuine and fraudulent transactions.
- To perform data cleaning and preprocessing.
- To conduct exploratory data analysis.
- To implement machine learning classification algorithms.
- To evaluate the performance of the developed models.

2. DATASET DESCRIPTION

The dataset used in this project is obtained from Kaggle's Credit Card Fraud Detection dataset.

Dataset overview:

Attribute	Description
Total records	284,807
Total Features	30
Time	Time elapsed between transactions
V1–V28	Anonymized numerical features
Amount	Transaction amount
Class	Target Variable

The target variable, Class, contains two categories:

- 0 – Genuine transaction
- 1 – Fraudulent transaction

The dataset contains a significant imbalance between genuine and fraudulent transactions, which is an important consideration during model development.

3. METHODOLOGY

The project follows a structured machine learning workflow consisting of the following stages:

1. Dataset collection.
2. Data inspection and cleaning.
3. Exploratory data analysis.
4. Feature selection.
5. Train-test splitting.
6. Feature scaling.
7. Machine learning model development.
8. Model evaluation.

4. WORK COMPLETED

4.1 Data Inspection and Cleaning

The dataset was initially inspected to understand its structure, dimensions, data types, and overall quality.

The following activities were performed:

- Check dataset dimensions and column names.
- Examined data types.
- Check for missing values.
- Identified duplicate records.
- Examined unique values.
- Performed outlier analysis.

- Validated the target variable.

The initial data cleaning and inspection stage have been completed.

4.2 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics and distribution of the transaction data.

The following analyses were conducted:

- Genuine versus fraudulent transaction distribution.
- Transaction amount analysis.
- Fraudulent transaction amount analysis.
- Genuine transaction amount analysis.
- Time-based analysis.
- Correlation analysis.

These analyses helped identify the imbalance between the two transaction classes and understand the distribution of transaction amounts.

4.3 Feature Preparation and Scaling

The dataset was divided into independent variables (X) and the dependent variable (y). The target variable was defined as Class, while the input features were prepared for classification.

The dataset was divided into training and testing subsets using the train_test_split function. StandardScaler was subsequently applied to standardize the numerical input features.

5. CURRENT PROJECT STATUS

Task	Status
Dataset collection	Completed
Data inspection and cleaning	Completed
Exploratory data analysis	Completed
Feature preparation	Completed
Train-test splitting	Completed
Feature scaling	Completed
Logistic Regression	Completed
Model evaluation	In progress
Model comparison	Pending

6. WORK IN PROGRESS

The current stage focuses on implementing Logistic Regression for binary classification. The model is being developed to predict whether a credit card transaction is genuine or fraudulent based on the available input features.

After model implementation, its performance will be evaluated using appropriate classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix. Further classification algorithms may also be implemented to compare their performance.

7. FUTURE WORK

The following activities are planned for the next stages:

1. Evaluate the model using classification metrics.
2. Implement additional classification algorithms such as Decision Tree and Random Forest.
3. Investigate techniques such as SMOTE to address class imbalance.
4. Compare model performance.
5. Interpret the final results and identify the most suitable model.

8. PRELIMINARY OBSERVATIONS

The initial exploratory analysis indicates that the dataset contains a substantial imbalance between genuine and fraudulent transactions.

This suggests that accuracy alone may not be sufficient to evaluate the classification model.

Precision, recall and F1-score will therefore be considered important evaluation of metrics during the upcoming model assessment stage.

9. REFERENCES

1. Kaggle – Credit Card Fraud Detection Dataset: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
2. Scikit-learn Documentation: https://scikit-learn.org/
3. Pandas Documentation: https://pandas.pydata.org/docs/
4. Matplotlib Documentation: https://matplotlib.org/

Note: This is an interim report. The machine learning implementation, model evaluation, and final conclusions will be updated as the project progresses.